

Agentic Engineering In Unreal

Let's tame the beast!

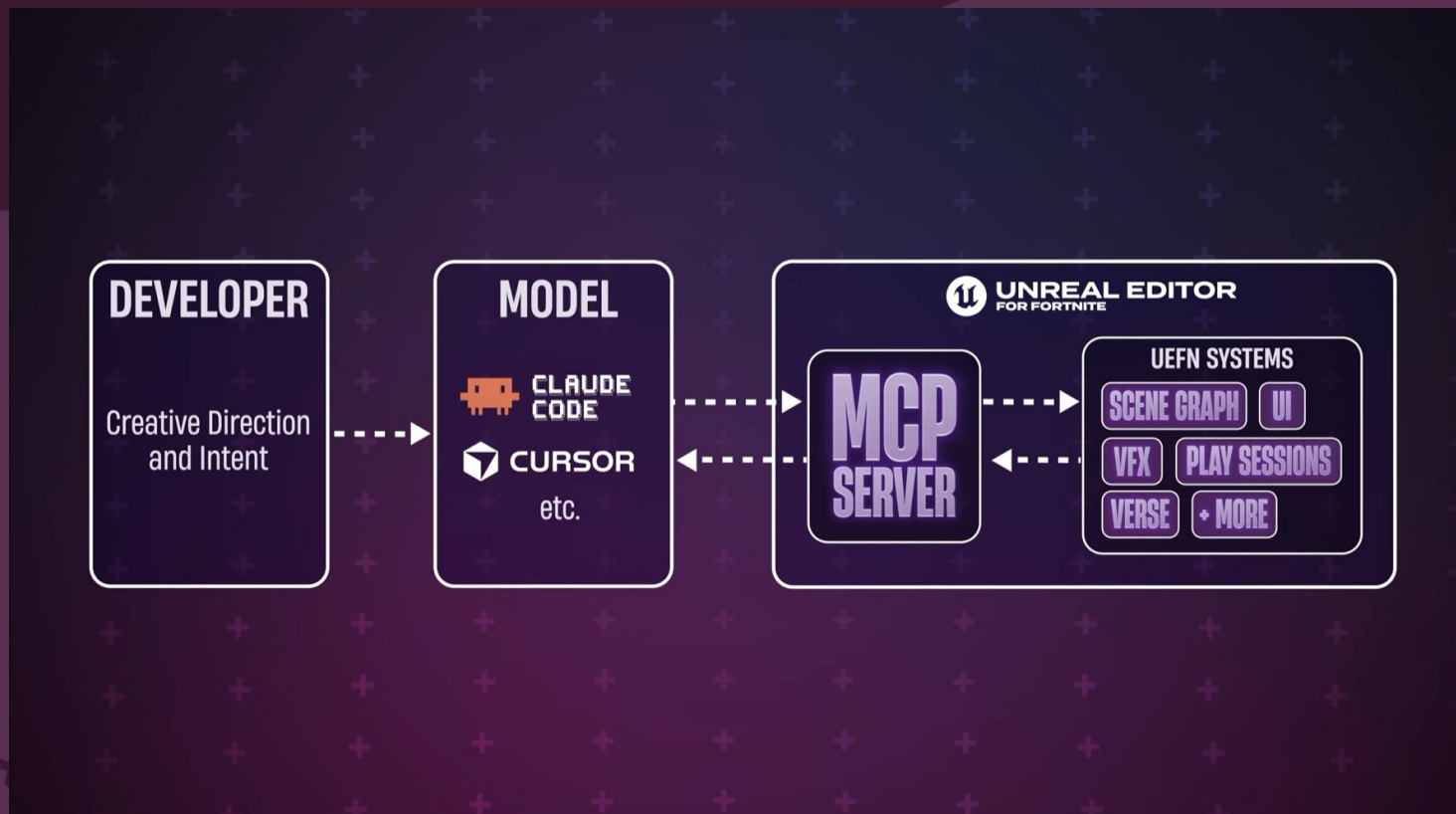




Why are we using a game engine again!?

- Control and Fast Iteration
 - High-level tooling (animation, NPC AI)
- Awesome features: Nanite, Lumen, Shaders, etc
 - Distribution Portability (*mostly free*)
 - Not reinventing the wheel

Epic's Vision for the future:



What's “Input and Output Harness”?



User harness

Feedforward and feedback controls put in place by the user of the coding agent for their particular system

Harness built by the builders of the coding agent, e.g. system prompt, code search tools, orchestration, ...

Model

Coding agent or SDK
E.g. Claude Code

How will we set up our project?

1. Create the Unreal project
2. Create the GIT repository pointing to it
3. Commit to GitHub *(or GitLab, we'll use GitHub in this bootcamp)*
4. Import the assets (models, sounds) and set up the main game scene
5. Set up the UE MCP with the agent UI *(Prefer a paid model provider if you can)*
6. Set up the User Harness Skills we'll use:
 - a. **Input Harness**
 - i. Coding style (C++)
 - ii. Game Design *(Compact a grill-me session to a skill)*
 - iii. Architectural Decisions *(Scene structure goes here as well)*
 - iv. Workflow *(Branches or worktrees, Loop or Gauntlet Loop)*
 - b. **Output Harness**
 - i. Code validation checks *(Be minimalistic, skip with advanced model)*
 - ii. Commit to repository
 - iii. Test the outcome *(Better with computer-use capabilities)*
 - iv. Create PR if has confidence on quality



Agentic Coding Strategy

Architecture

Implementation

Top-down:

From high-level
vision to fine details



Bottom-Up:

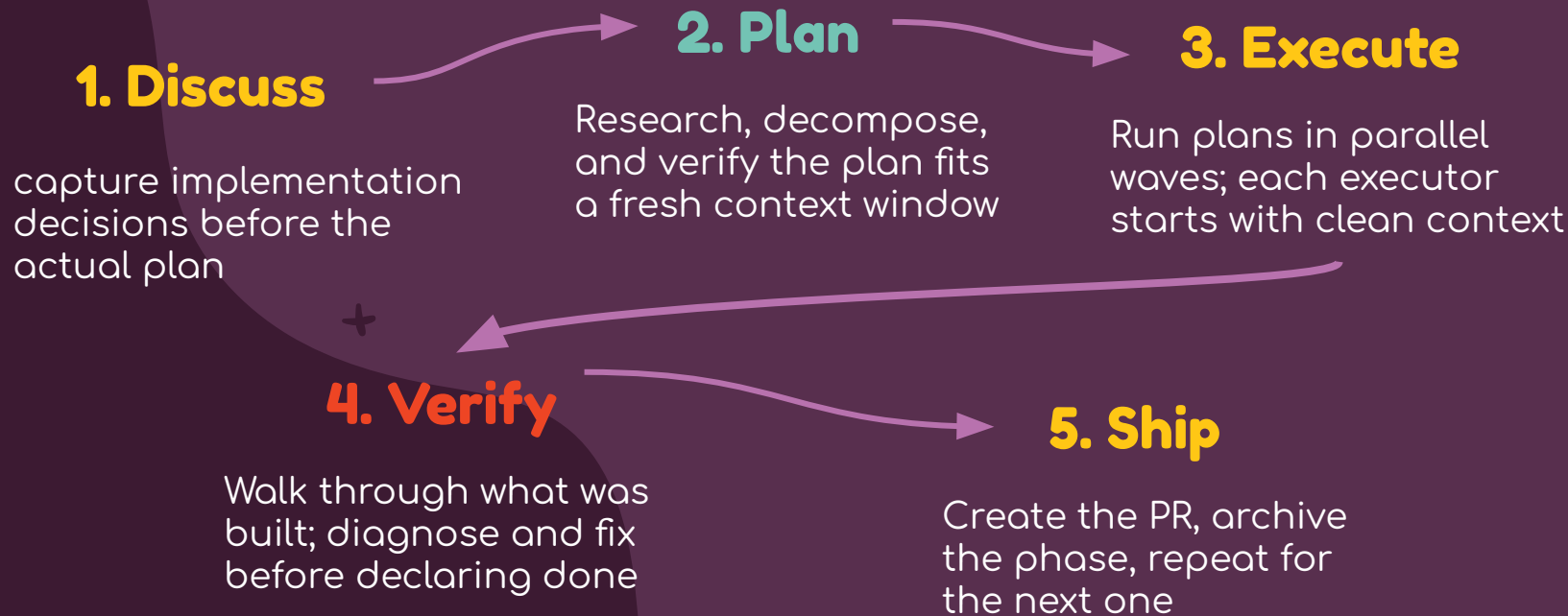
From low-level
components to
high-level
managers



The GSD Skill

<https://github.com/open-gsd/gsd-core>

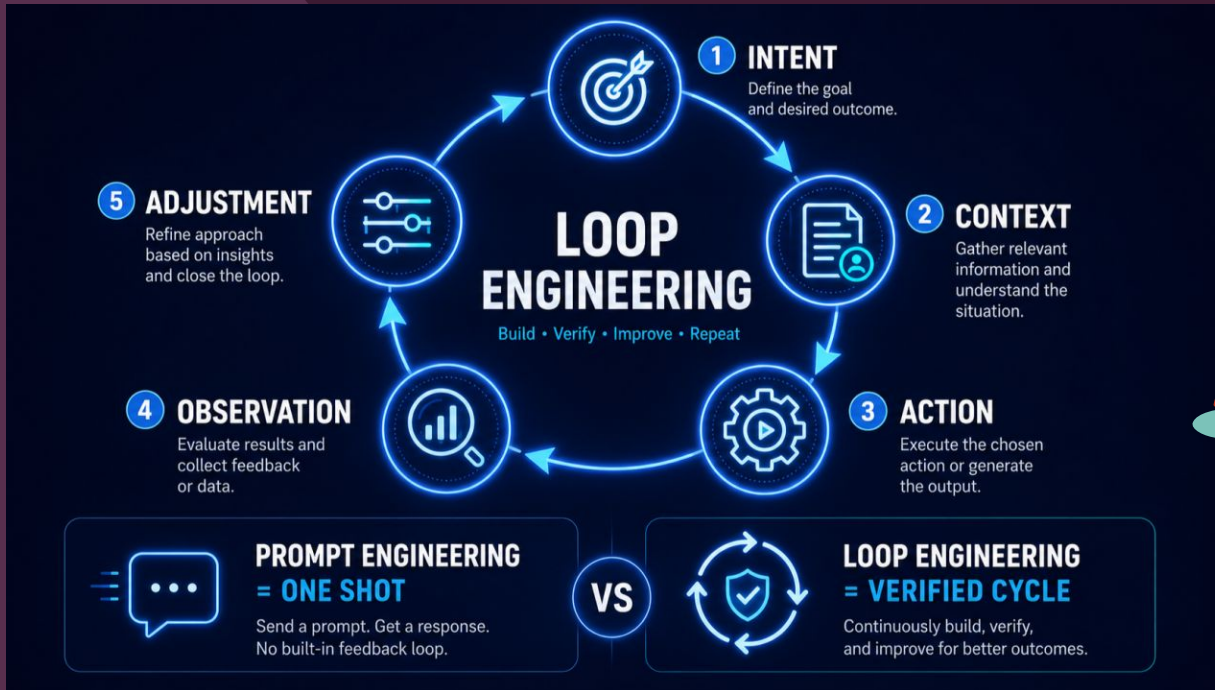
aka “Get Sh.. tuff done”



Self-improving Agentic Strategies

1. Loop Engineering

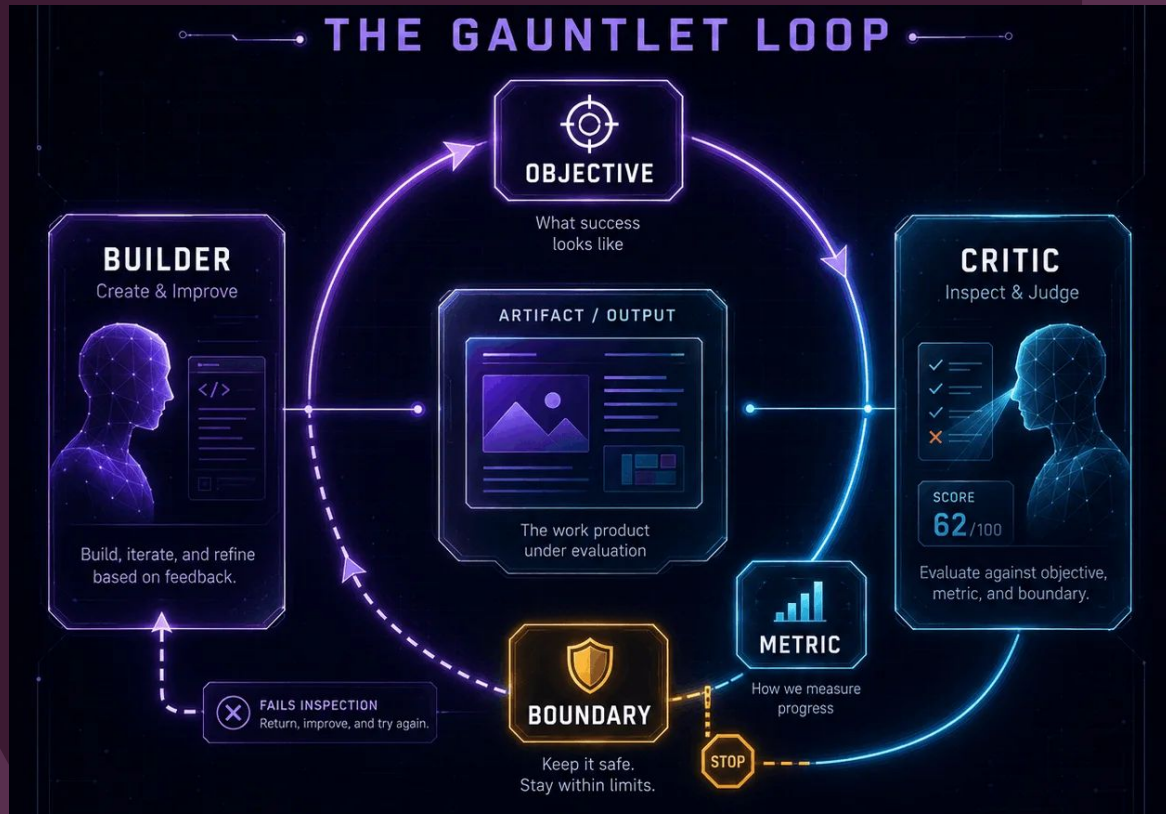
<https://github.com/vibhasdutta/loop-engineer>

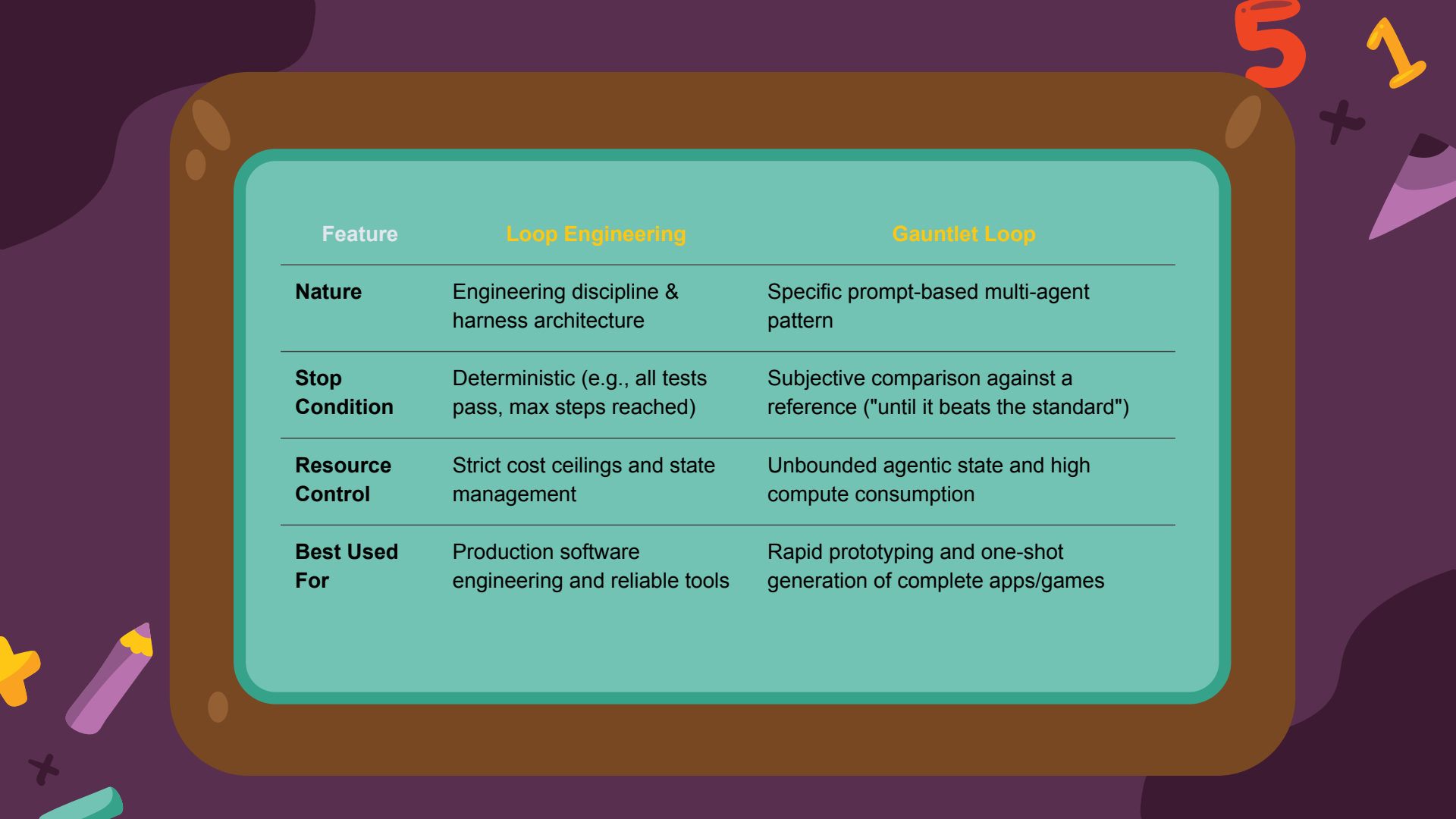


Self-improving Agentic Strategies

2. Gauntlet Loop

<https://github.com/robonuggets/gauntlet-loop>





The background of the slide is a dark purple color. In the top right corner, there is a large orange number '5' and a yellow number '1' with a black plus sign between them. In the bottom left corner, there is a yellow plus sign and a purple pencil. In the bottom right corner, there is a purple pencil and a black plus sign.

Feature	Loop Engineering	Gauntlet Loop
Nature	Engineering discipline & harness architecture	Specific prompt-based multi-agent pattern
Stop Condition	Deterministic (e.g., all tests pass, max steps reached)	Subjective comparison against a reference ("until it beats the standard")
Resource Control	Strict cost ceilings and state management	Unbounded agentic state and high compute consumption
Best Used For	Production software engineering and reliable tools	Rapid prototyping and one-shot generation of complete apps/games

Questions?

